Swiss Personalized Health Network: Making health data FAIR in Switzerland

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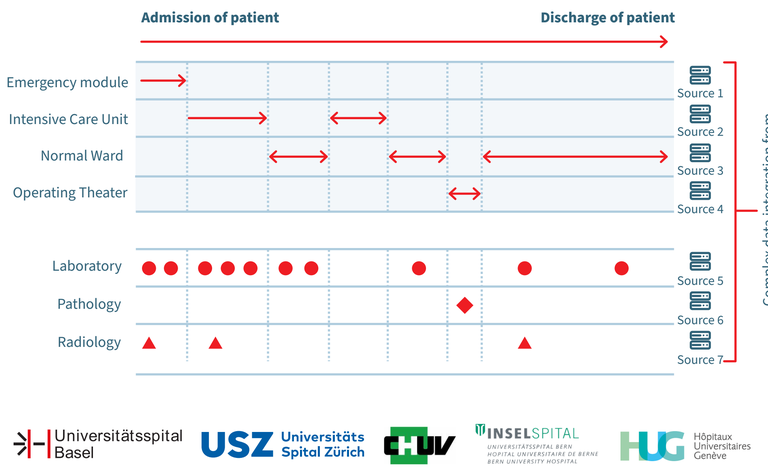
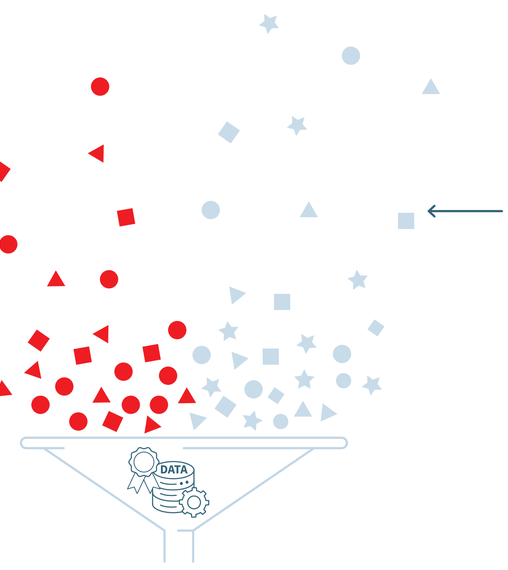
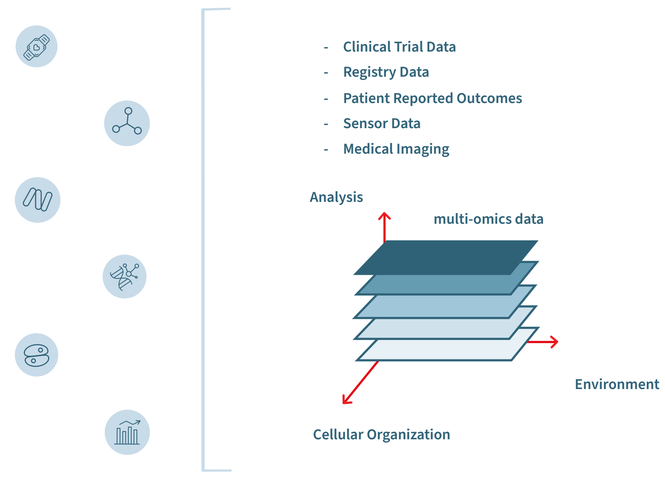
Documentation GitLab Training

The Swiss Personalized Health Network (SPHN) is a national initiative under the leadership of the Swiss Academy of Medical Sciences (SAMS). In collaboration with the SIB Swiss Institute of Bioinformatics, the SPHN contributes to the development, implementation, and validation of coordinated data infrastructures in order to make clinical and health-related data available for research in Switzerland. Since future research questions and their data requirements are constantly evolving, the priority was to develop a purpose-independent description of existing data and knowledge rather than relying on existing data models which are typically focused on a specific use case. Together with partners at University Hospitals we have developed the SPHN Semantic Interoperability Framework that enables collaborative research by turning routine health data into FAIR (Findable, Accessible, Interoperable, Reusable) research data. The framework is built with a three pillar strategy that focuses on the definition of semantics for data unambiguity and standardization, data format specifications for data exchange, and explore use-cases and interoperability with external data models and initiatives. The framework also includes software tools and services to automate and facilitate the adoption of the semantics and data exchange.







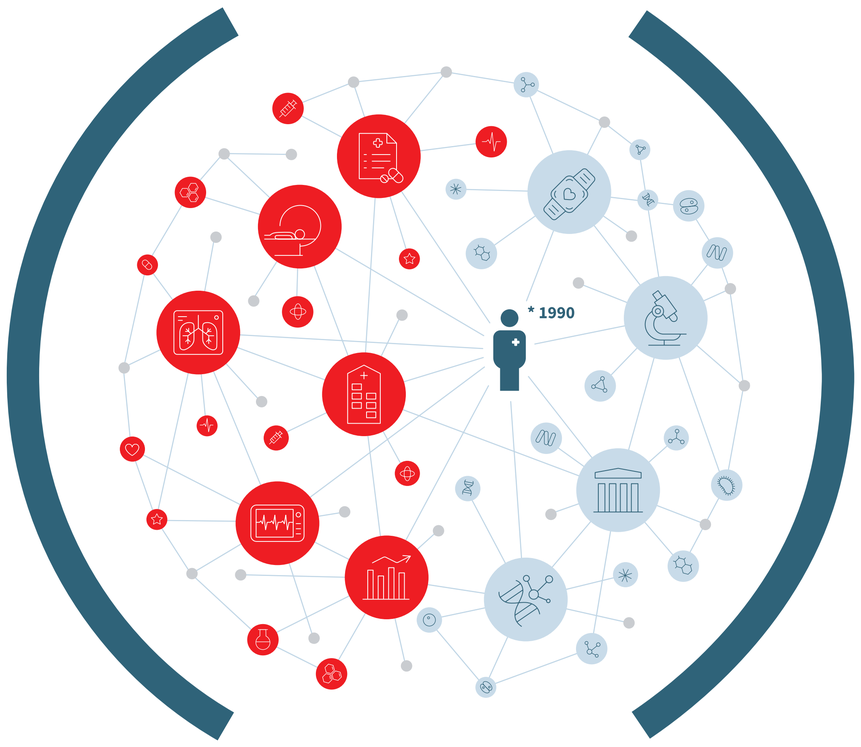
Semantics

The semantics are captured as concepts and attributes that can be used as building blocks to define more complex concepts.

Oxygen Saturation	fraction of oxygen present in the blood	SNOMED:103228002
quantity	value and unit of the concept	
measurement datetime	datetime of measurement	
measurement method	method of measurement of the concept	
body site	body site where the concept was measured, performed or collected	

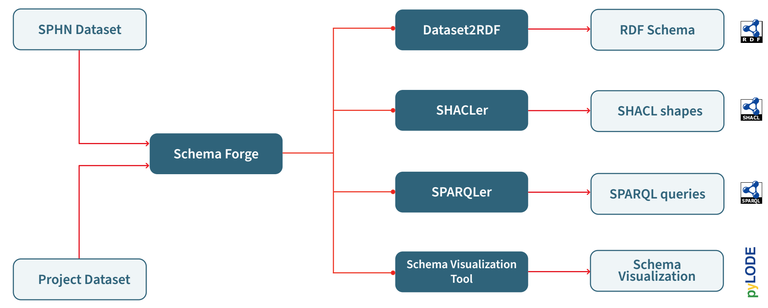
SNOMED CT LOINC

The **SPHN Dataset** is the collection of concepts, attributes, meaning binding, value sets and subsets, and any additional metadata that enriches the semantics.



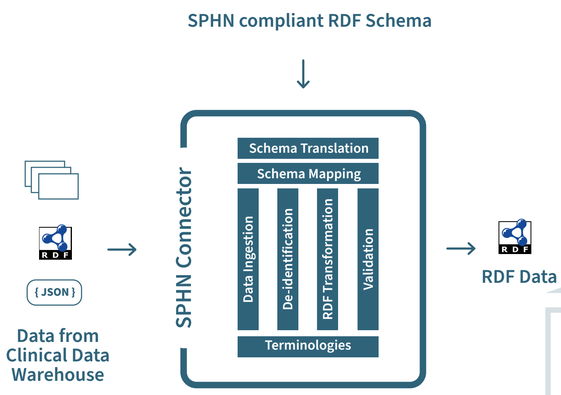
Knowledge Representation

The collection of concepts and attributes are then modeled using RDFS, SKOS and OWL, and represented in RDF.



The **SPHN Schema Forge** is a web service that translates SPHN or SPHN Project Dataset into its corresponding RDF Schema, SHACL shapes, SPARQL queries, and schema visualization.

Data Preparation



Data providers, such as university hospitals, make use of the defined semantics and its corresponding formal representations to prepare the data.

The **SPHN Connector** is a tool that facilitates the creation and validation of RDF data from typical stores like relational databases, CSV, and JSON. The tool also provides the ability to perform quality checks on the data and also apply certain de-identification strategies as part of the data transformation.

The tool serves as a easy entry point for new data providers to map their data onto the SPHN RDF Schema and transform to RDF.

Terminology Service

The DCC Terminology Service provides SPHN compatible, machine readable representations of national and international terminologies that are important in the context of SPHN.

The terminologies are provided in RDF without external dependencies and in compliance with copyright statements of the terminologies.

Since 2023, the DCC has adopted the concept of versioning to enhance interoperability of codes from different versions of a terminology.

Following are a list of terminologies that are FAIRified, translated to RDF, and provided by the DCC Terminology Service:

- **ATC:** Anatomical and Therapeutic Chemical Classification
- **SNOMED-CT:** Systematized Nomenclature of Medicine Clinical Terms
- **CHOP:** Schweizerische Operationsklassifikation
- **ICD-10-GM:** International Statistical Classification of Diseases, 10th Revision, German Modification
- **LOINC:** Logical Observation Identifier Names and Codes
- **UCUM:** Unified Code for Units of Measure
- **HGNC:** HUGO Gene Nomenclature

Projects

Driver Projects

SFNR: Swiss Frailty Network and Repository

SPO: Swiss Personalized Oncology

PSSS: Personalized Swiss Sepsis Study

IMAGINE: Radiomics for comprehensive patient and disease phenotyping in personalized health

National Data Streams

SPO-NDS: Swiss Personalized Oncology National Data Stream

SwissPedHealth: Pediatric personalized research network Switzerland

LUCID: Low Value of Care in Hospitalized Patients

IICU: Personalized, data-driven prediction and assessment of infection-related outcomes in Swiss ICUs

Demonstrators

SwissPedDW: Swiss Pediatric Data Warehouse

ADONIS: Accelerating detection of neonatal sepsis

RDeID: Risk-based de-identification platform for health-related data

SNOW: Swiss Network of Wearables

S4: Smart SNOMED Search for SPHN

PREPP: PReparing PERsonalizEd PEdiatRic PRimaRy caRE

SPHN SPAC: Using routine health care data to facilitate clinical cohort studies

Acknowledgement and References

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SIB Swiss Institute of Bioinformatics

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